

Carrier Recovery

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1 Introduction

The performance and energy consumption of several combinations of frequency and phase recovery algorithms are compared. CORDIC will be used to find angles and approximated as 3n additions. Direct complex multiplication used to apply phase rotations. Bit shifts will be assumed to have negligible energy cost.

2 Frequency Recovery

2.1 Isolating CFO

All frequency recovery algorithms first require that the phase of the data be removed before estimation. For QPSK symbols $c[i] = e^{j\frac{\pi}{4} + m[i]\frac{\pi}{2}}$ this can be done by raising each received symbol $x[i] = c[i]e^{j(2\pi f_d i + \theta[i])} + n[i]$ to the fourth power to give

$$z[i] = x^4[i] = e^{j(8\pi f_d i + 4\theta[i])} + n'[i] \quad (1)$$

where θ is phase noise, f_d is the CFO, and n is additive noise. This allows all data symbols to be used in CFO estimation, but amplifies phase and additive noise which degrades the estimation quality. The fourth power operation also requires three complex multiplications. An alternative is to use pilot symbols where the data phase is known beforehand, giving

$$z[i] = x[i]c^*[i] = e^{j(2\pi f_d i + \theta[i])} + n[i], \quad (2)$$

which is more robust to noise and less computationally expensive. Some algorithms only require $\arg(z[i])$ which is efficiently found using CORDIC as

$$\arg(z[i]) = \arg(x[i]) - \arg(c[i]). \quad (3)$$

2.2 Rife and Boorstyn (FFT Search)

The value of the CFO will appear as the strongest peak in the frequency spectrum of $z[i], z[i+1], \dots, z[i+L]$ where L is the observation length. The Rife and Boorstyn algorithm first performs an FFT on the observed sequence and finds the index k corresponding to the strongest peak. This coarse estimation is refined with a parabolic interpolation on the complex FFT values X_{k-1}, X_k, X_{k+1} to give

$$\delta = -Re \left[\frac{X_{k+1} - X_{k-1}}{2x_k - X_{k-1} - X_{k+1}} \right]. \quad (4)$$

The updated index $k' = k + \delta$, which is then scaled by the sampling rate to find the true CFO. The CPON specification allows for 11 training symbols at the start of each subframe. The resolution in k is determined by the FFT size N_{FFT} as $\epsilon(k) = 1/2N_{FFT}$. For data-aided operation, only the 11 training symbols are available. This would give a resolution of $\epsilon(k) \approx 0.05$. In terms of the estimated frequency $\hat{f}_d$ and symbol rate R_s , $\epsilon(\hat{f}_d) \approx 0.05R_s$. This would likely be too large for the downstream phase recovery to correct, so a larger FFT can be used with zeros padded at the end of the observed sequence. The normalised mean squared-error (MSE) in the estimated CFO is shown in fig. 1 for data-aided operation with 11 training symbols. A CFO estimate is computed for both polarizations and averaged to reduce the MSE. The modified Cramer-Rao bound (MCRB) gives a lower bound on the achievable MSE and is given by

$$MCRB = \frac{3}{2\pi^2 L^3 \times SNR}. \quad (5)$$

The Rife and Boorstyn achieves this bound for SNRs over ≈ 4 dB. At lower SNRs, the noise has a significant contribution to the FFT values and can lead to an incorrect index being chosen.

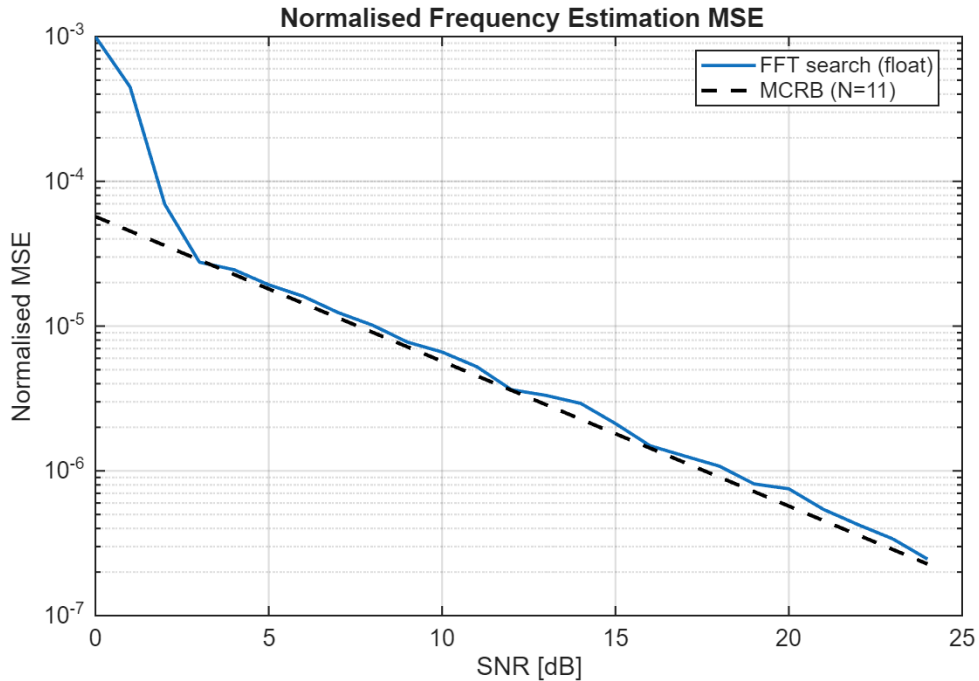


Figure 1: Normalised MSE of the Rife and Boorstyn algorithm with FFT size $N_{FFT} = 512$ compared to the MCRB for $N = 11$ training symbols.

Data-aided operation is constrained by the length of the training sequence but the accuracy of the Rife and Boorstyn algorithm improves with longer observation lengths, as shown in fig. 2 for non data-aided operation. The SNR threshold at which this algorithm begins to deviate from the MCRB is much higher with data aided operation due to the amplification of noise as seen in eq. (1). Combined with the increased computational cost of removing the data phase in non data-aided operation, this makes the use of training symbols preferable so long as the training sequence is not too short.

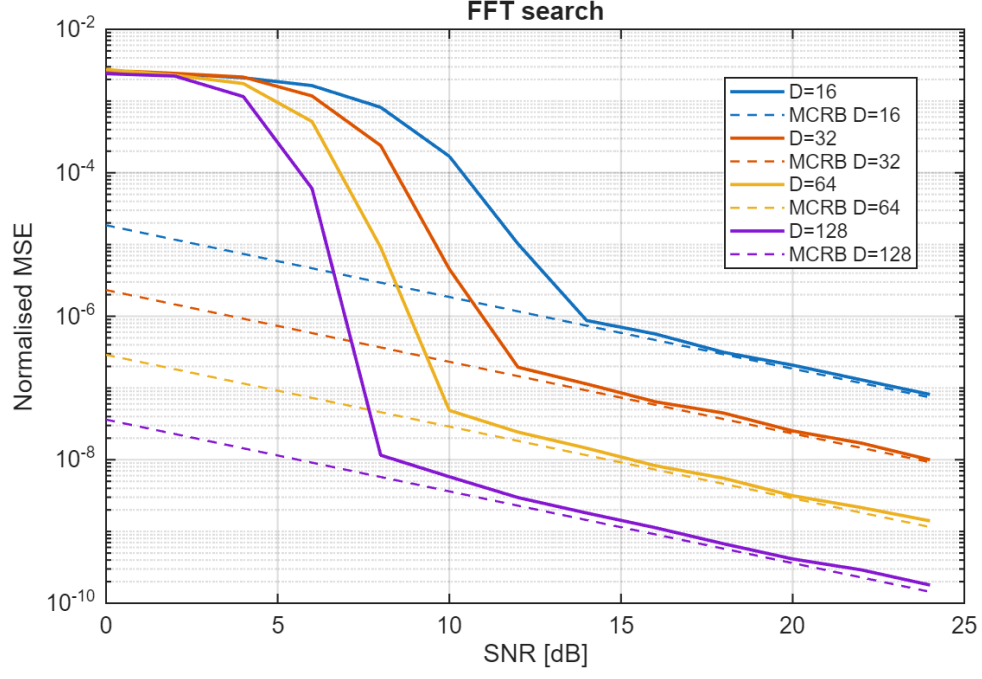


Figure 2: Normalised MSE of the Rife and Boorstyn algorithm with FFT size $N_{FFT} = 512$ compared to the MCRB for $N = 11$ training symbols.